

GE23131-Programming Using C-2024

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Status	Finished
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Duration	1 hour 22 mins

Question 1

Correct

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Coders here is a simple task for you, you have given an array of size **N** and an integer **M**.

Your task is to calculate the **difference between maximum sum and minimum sum of N-M** elements of the given array.

Constraints:

$1 \leq t \leq 10$
 $1 \leq n \leq 1000$
 $1 \leq a[i] \leq 1000$

Input:

First line contains an integer **T** denoting the number of testcases.
First line of every testcase contains two integer **N** and **M**.

Output:

For every test case print your answer in new line

SAMPLE INPUT

```
1
5 1
1 2 3 4 5
```

SAMPLE OUTPUT

```
4
```

Explanation

M is 1 and N is 5 so you have to calculate maximum and minimum sum using $(5-1 =)$ 4 elements.

Maximum sum using the 4 elements would be $(2+3+4+5=)$ 14.

Minimum sum using the 4 elements would be $(1+2+3+4=)$ 10.

Difference will be $14-10=$ 4.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
```

```
6 while(t--)\n7 {\n8     int n,m,d,min,temp;\n9     scanf("%d %d",&n,&m);\n10    d=n-m;\n11    int arr[n];\n12    for(int i=0;i<n;i++)\n13        scanf("%d",&arr[i]);\n14    for(int j=0;j<n;j++)\n15    {\n16        min=j;\n17        for(int k=j;k<n;k++)\n18        {\n19            if(arr[k]<arr[min])\n20                min=k;\n21        }\n22        temp=arr[min];\n23        arr[min]=arr[j];\n24        arr[j]=temp;\n25    }\n26    int maxsum=0,minsum=0;\n27    for(int a=0;a<d;a++)\n28        minsum+=arr[a];\n29    for(int b=n-1;b>m-1;b--)\n30        maxsum+=arr[b];\n31    printf("%d\\n",maxsum-minsum);\n32 }\n33 \n34
```

	Input	Expected	Got	
✓	1 5 1 1 2 3 4 5	4	4	✓

Question **2**

Correct

Marked out of
1.00[Flag
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A new deadly virus has infected large population of a planet. A brilliant scientist has discovered a new strain of virus which can cure this disease. Vaccine produced from this virus has various strength depending on midichlorians count. A person is cured only if midichlorians count in vaccine batch is more than midichlorians count of person. A doctor receives a new set of report which contains midichlorians count of each infected patient, Practo stores all vaccine doctor has and their midichlorians count. You need to determine if doctor can save all patients with the vaccines he has. The number of vaccines and patients are equal.

Input Format

First line contains the number of vaccines - N. Second line contains N integers, which are strength of vaccines. Third line contains N integers, which are midichlorians count of patients.

Output Format

Print a single line containing '**Yes**' or '**No**'.

Input Constraint

$$1 < N < 10$$

Strength of vaccines and midichlorians count of patients fit in integer.

5

123 146 454 542 456

100 328 248 689 200

SAMPLE OUTPUT

No

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      int n,a,b,temp,flag=1;
5      scanf("%d",&n);
6      int vac[n],pat[n];
7      for(int i=0;i<n;i++)
8          scanf("%d",&vac[i]);
9      for(int i=0;i<n;i++)
10         scanf("%d",&pat[i]);
11     for(int j=0;j<n-1;j++)
12     {
13         a=j,b=j;
14         for(int k=j;k<n;k++)
15         {
16             if(vac[k]<vac[a])
17                 a=k;
18             if(pat[k]<pat[b])
19                 b=k;
20         }
21
22         temp=vac[a];
```

```
26     temp=pat[b];
27     pat[b]=pat[j];
28     pat[j]=temp;
29 }
30 for(int i=0;i<n;i++)
31 {
32     if(vac[i]<=pat[i])
33     {
34         flag=0;
35         break;
36     }
37 }
38 if(flag==1)
39     printf("YES");
40 else
41     printf("No");
42 }
```

	Input	Expected	Got	
✓	5 123 146 454 542 456 100 328 248 689 200	No	No	✓

Passed all tests! ✓

Question **3**

Correct

Marked out of 1.00

You are given an array of n integer numbers $a_1, a_2, \dots, a_n$. Calculate the number of pair of indices (i, j) such that $1 \leq i < j \leq n$ and $a_i \text{ xor } a_j = 0$.

Input format



- Second line: n space separated integers $a_1, a_2, \dots, a_n$.

Output format

Output the required number of pairs.

Constraints

$$1 \leq n \leq 10^6$$

$$1 \leq a_i \leq 10^9$$

SAMPLE INPUT

5

1 3 1 4 3

SAMPLE OUTPUT

2

Explanation

The 2 pair of indices are **(1, 3)** and **(2,5)**.

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int a,c=0;
5     scanf("%d",&a);
6     int arr[a];
7     for(int i=0;i<a;i++)
8         scanf("%d",&arr[i]);
9     for(int i=0;i<a;i++)
10    {
11        for(int j=i+1;j<a;j++)
12        {
13            if((arr[i]^arr[j])==0)
14                c++;
15        }
16    }
17     printf("%d",c);
18 }
```

	Input	Expected	Got	
✓	5 1 3 1 4 3	2	2	✓

Passed all tests! ✓

Marked out of
1.00

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question](#)

array.

Example:

$A=\{4,5,3,7,1\}$

After sorting the new array becomes $A=\{1,3,4,5,7\}$.

The required output should be "4 2 0 1 3"

INPUT :

The first line of input consists of the size of the array

The next line consists of the array of size m

OUTPUT :

Output consists of a single line of integers

CONSTRAINTS:

$1 \leq m \leq 106$

$0 \leq A[i] \leq 106$

NOTE: The indexing of the array starts with 0.

5

4 5 3 7 1

SAMPLE OUTPUT

4 2 0 1 3

Answer: (penalty regime: 0 %)

```
1 #include<stdio.h>
2 int main()
3 {
4     int n;
5     scanf("%d",&n);
6     int arr[n];
7     for(int i=0;i<n;i++)
8         scanf("%d",&arr[i]);
9     int max=arr[0];
10    for(int i=1;i<n;i++)
11    {
12        if(arr[i]>max)
13            max=arr[i];
14    }
15    max++;
16    int min=0;
17    for(int a=0;a<n;a++)
18    {
19        for(int b=0;b<n;b++)
20        {
21            if(arr[b]<arr[min])
22                min=b;
23        }
24        printf("%d ",min);
25        arr[min]=max;
26    }
```

	Input	Expected	Got	
✓	5 4 5 3 7 1	4 2 0 1 3	4 2 0 1 3	✓

Passed all tests! ✓

Finish review

